NUI Galway
OÉ Gaillimh

Semester 2 Examinations 2021/2022

Exam Code(s)	3BA1, 3BMS2, 3BS9, 4BCS1, 4BFS1
Exam(s)	Bachelor of Science (Hons.) Bachelor of Science (Mathematical Science) Honours Bachelor of Arts
Modules	Fields and Applications
Module Codes	MA3491
Paper	1
External Examiner(s)	Prof. C. Roney-Dougal
Internal Examiner(s)	Dr N. Madden ⋆ Dr E. Sköldberg
Instructions:	**Students should attempt ALL questions.**
Duration	2 Hours
No. of Pages	3 (including this page)
Discipline	Mathematics

Requirements:

Release in exam venue	Yes ☑ No ☐
Release to Library	Yes ☑ No ☐
Handout	None
Log Tables	None
Cambridge Tables	None
Graph Paper	None
Other materials	Non-programmable calculators are permitted

1. (a) State the *sphere packing bound*, and determine which upper bound on the number of codewords in a block code with alphabet $\mathbb{F}_5$ of block length 8 that corrects all single errors it gives. **[8 marks]**

   (b) In the following ISBN numbers, one symbol has been replaced by a '`#`'. Find the missing symbol.

      (i) `1-56639-404-#` **[3 marks]**

      (ii) `978-#-87548-591-2` **[3 marks]**

   (c) Consider the code C over the alphabet $\{0, 1, 2\}$ which consists of all words of length 9 where every symbol occurs exactly three times. Determine the minimum distance of C. **[6 marks]**

2. (a) For each of the following polynomials in $\mathbb{F}_3[x]$, determine if it is irreducible or not, and if it is reducible, factor it into irreducibles.

      (i) $x^4 + x^2 + 1$. **[3 marks]**

      (ii) $x^3 + 2x + 1$. **[3 marks]**

   (b) Show that $f(x) = x^3 + x^2 + 1 \in \mathbb{Z}_2[x]$ is irreducible, and hence let $\mathbb{F}_8 = \mathbb{Z}_2[x]/(f(x))$. **[2 marks]**

   (c) Calculate the product $(x^2 + 1)(x^2 + x + 1)$ in $\mathbb{F}_8$. (Give the result as the residue class of a polynomial of degree ≤ 2). **[6 marks]**

   (d) Calculate the multiplicative inverse of x^2 in $\mathbb{F}_8$. (Give the result as the residue class of a polynomial of degree ≤ 2). **[6 marks]**

P.T.O

3. (a) Let C be the linear code over the field $\mathbb{F}_2$ with generator matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

(i) Write down the code words of C. **[4 marks]**

(ii) Determine the minimum distance of C. **[4 marks]**

(b) Let C be the code over $\mathbb{F}_4$, the field with four elements, with generator matrix

$$\begin{pmatrix} 01 & 01 & 00 & 00 & 01 & 01 \\ 01 & 00 & 01 & 01 & 00 & 01 \\ 00 & 01 & 01 & 01 & 01 & 00 \end{pmatrix}.$$

(i) Find a generator matrix in standard form for C. **[4 marks]**

(ii) Hence, construct a parity check matrix for C. **[4 marks]**

(iii) Hence, determine if the word `10 01 00 11 10 01` is a codeword in C. **[4 marks]**

4. (a) (i) What is the definition of the Hamming code $Ham(r, q)$? **[2 marks]**

(ii) Write down a parity check matrix for the Hamming code $Ham(2, 5)$. **[3 marks]**

(iii) Explain how a single error can be corrected in $Ham(2, 5)$. Decode the words `123321` and `102030`. **[4 marks]**

(b) Let C be the cyclic code of length 7 over the alphabet $\mathbb{F}_2$ with generator polynomial $x^4 + x^3 + x^2 + 1$.

(i) Give the generator matrix of C. **[3 marks]**

(ii) Find the check polynomial of C. **[4 marks]**

(iii) Write down the check matrix of C and hence determine if any of the words `0111001` or `0011011` is a codeword in C. **[4 marks]**